System Design Document

Airport Baggage Handling Data System

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1 Introduction

This document presents the design of a data-driven system for analyzing an airport baggage handling dataset. The dataset consists of sensor-based event logs capturing baggage movement across different processes such as check-in, screening, sorting, and transport.

The goal of the system is to provide visibility into operational performance, detect inefficiencies, and support decision-making through data analysis.

2 Data Sources

The system is based on the following sources:

- Dataset Description Document (system context, variables, and meaning) [1]
- Baggage Handling Dataset (CSV file with event logs and sensor data)[2]

The dataset contains time-series records with operational and environmental variables such as:

- Bag ID
- Timestamp
- Process (check-in, screening, sorting, transport)
- Sensor readings (temperature, vibration, speed)
- RFID Reading
- Result (success/failure)
- Priority flag
- Screen Delay time
- Vehicle Delay time

3 System Overview

The system transforms raw sensor data into insights through the following pipeline:

1. Data ingestion from CSV
2. Data cleaning and preprocessing
3. Storage in structured format
4. Analytical processing
5. Visualization and reporting

4 Entity-Relationship Diagram

Figure 1 outlines the database structure for a tracking system, likely for luggage or cargo in a transit setting.

The diagram illustrates how four main entities interact:

Bag: Stores information about individual pieces of luggage, including a unique ID, flight information, and whether it has priority status.

Event: Tracks specific occurrences related to a bag, recording the time, any delays, and whether the process was a success or failure.

Process: Defines the different locations or phases (e.g., check-in, screening) an event can belong to.

SensorReading: Captures physical data associated with an event, such as temperature, vibration, and speed, allowing for detailed monitoring of how bags are handled.

In summary, this schema is designed to provide a comprehensive log of a bag's journey by linking physical sensor data and status updates to specific bags and transit stages.

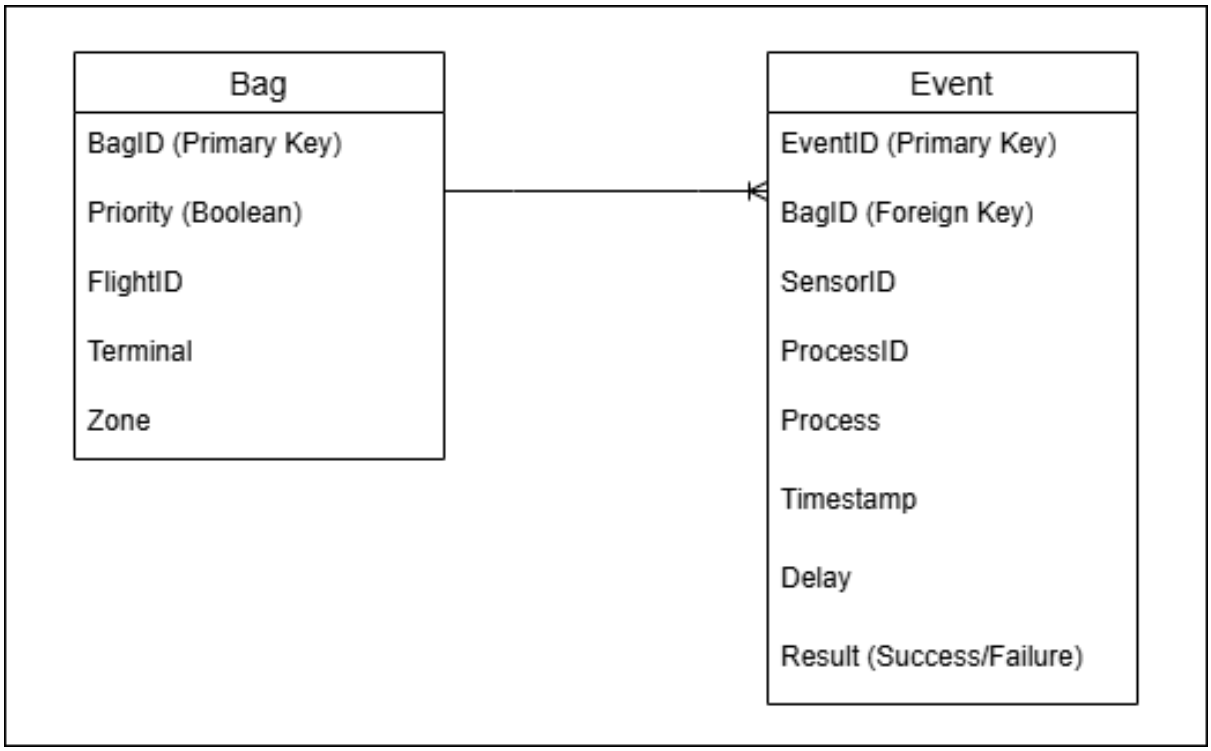

Figure 1: Entity-Relationship Diagram

5 System Architecture

Figure 2: System Architecture

6 Workflows

6.1 Data Pipeline Workflow

1. Load the CSV dataset
2. Clean missing or inconsistent values
3. Transform timestamps and categorical variables
4. Store structured data
5. Compute KPIs (delay, throughput, failure rate)

6.2 Baggage Handling System Flow

[1]

Figure 3: Baggage Handling System Workflow

7 Analytics and KPIs

The system computes the following metrics:

- Average delay per stage
- Bottleneck identification
- Failure rate per stage
- Sensor anomaly detection
- Priority handling performance

8 Design Considerations

8.1 Scalability

The system should support large datasets and potential real-time extensions.

8.2 Reliability

Sensor failures and missing data must be handled accordingly.

8.3 Data Quality

Data validation and cleaning are essential due to noisy sensor inputs.

8.4 Performance

Efficient querying is required for timely analysis.

9 Conclusion

This document presents a system design for analyzing baggage handling operations using sensor data. By structuring the dataset into a clear data model and implementing analytical workflows with easy to understand dashboard and reporting, the system enables identification of inefficiencies and supports operational improvements.

References

[1] *Airport Data Insights – Dataset Description Document.* data.docx

[2] *Baggage Handling Dataset (CSV).*dataset.csv